

# Weekly Progress Report

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**Domain:** Machine Learning / Industrial ML

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**Week Ending:** 02

## I. Overview

This week focused on exploratory data analysis, preprocessing, feature engineering, and baseline dataset preparation for both industrial ML projects.

## II. Achievements

1. Project 10: Analyzed distributions, missing values, correlations, temporal trends, and relationships between process variables and silica concentration.
2. Project 10: Investigated sampling-frequency differences and prepared a consistent time-based representation for modelling.
3. Project 10: Created initial lag, rolling-statistic, and time-derived features where appropriate.
4. Project 6: Performed engine-wise exploratory analysis of sensor degradation across operational cycles.
5. Project 6: Identified constant/low-information sensors and prepared cycle-based windows for supervised learning.
6. Documented preprocessing decisions so that the pipeline can be reproduced and extended.

## III. Challenges

1. Feature engineering for time-dependent data can introduce data leakage if future information enters the feature window.
2. Sensor readings contain noise and operating-condition effects that can obscure degradation patterns.
3. Choosing an appropriate resampling/windowing strategy requires balancing information retention and computational cost.

## IV. Learning Resources

1. pandas and NumPy documentation for time-series operations.
2. scikit-learn preprocessing and model-selection documentation.
3. Technical references on time-series feature engineering and predictive maintenance.

## V. Next Week's Goals

1. Train simple baseline models for both projects.
2. Establish evaluation metrics and validation strategies.
3. Compare baseline predictions against business/engineering objectives.
4. Begin model comparison and feature-importance analysis.

## VI. Additional Comments

The main outcome was a cleaner and reproducible data pipeline, providing a reliable foundation for model development.